Tenju Cuddihy

GEOSPATIAL DATA SCIENTIST

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PROFILE

Geospatial data scientist working with large, multi-source forest and environmental datasets. First author of a forthcoming FEMC regional report comparing regeneration and overstory composition across more than 10,000 monitoring plots in the Northeast, built on R pipelines I wrote to standardize and analyze the underlying inventory data. I also build automated Python geoprocessing workflows, and since 2023 have worked remotely and largely independently, turning inventory and remote-sensing records into analysis-ready results for foresters, land managers, and researchers.

EXPERIENCE

Data Scientist / Analyst

Mar 2026 – Present · Remote

Forest Ecosystem Monitoring Cooperative (FEMC), University of Vermont

- First-authored a forthcoming FEMC regional report, *Northeastern Forest Regeneration: A Statistical Analysis*, comparing overstory and regeneration composition across **10,632 forest monitoring plots** in the Northeastern U.S. (2015–2024), using compositional methods (Bray-Curtis dissimilarity, Shannon diversity) to flag the forest types where sapling and seedling layers diverge most from the current canopy.
- Built version-controlled (Git) R pipelines that standardized and harmonized forest inventory data from three independent monitoring networks (FIA, FHM, NEFIN), each with its own sampling design, species conventions, and plot layout, into a single analysis-ready dataset.
- Developed an automated Python pipeline (geopandas, rasterio, GDAL, SciPy) generating **30 annual 10 m distance-to-infrastructure rasters** (roads, buildings, and combined infrastructure, 2016–2025) for the **Northeastern U.S. (7 states, ~200 counties)**, built from infrastructure data spanning nine states plus adjacent Canada so borders don't distort distances, as inputs to a forest-disturbance model automating TIGER/Line and OpenStreetMap retrieval with QA against Microsoft Building Footprints.

Geospatial Technician

Mar 2023 – Dec 2025 · Burlington, VT, then remote

Spatial Analysis Laboratory (SAL), University of Vermont

- Classified and quality-reviewed high-resolution land cover (tree canopy, impervious surface, buildings, and road networks) across **18 mapping projects in 15+ US states and Canada**, logging roughly **2,760 hours in ArcGIS Pro between 2023 and 2025**.
- Delivered full land-cover classification and multi-pass quality review for large regional programs, including the **Chesapeake Bay Watershed (MD/VA), EPA Area 3 (PA/VA), Texas statewide, the California Central Coast, and New York City**.
- Processed LiDAR point clouds in QTM and specialized in urban tree-canopy and tree-canopy-change mapping supporting land-cover monitoring and urban-planning applications.

Research Assistant

Jan 2022 – May 2023 · Burlington, VT

Nutrient Recycling & Ecological Design Lab, University of Vermont

- Measured phosphorus cycling across **five restored wetland sites** using lab methods (TSS/TVS, loss-on-ignition) and field-deployed YSI multiparameter sensors, logging water temperature, dissolved oxygen, and water level at 10-minute intervals (Feb–May 2022).
- Authored an undergraduate honors thesis (see Publications) on dissolved-oxygen dynamics and what they imply for phosphorus cycling in the Lake Champlain Basin.

PUBLICATIONS

Cuddihy, T., & Donisvitch, S. (2026, in press). *Northeastern Forest Regeneration: A Statistical Analysis.* Forest Ecosystem Monitoring Cooperative (FEMC), University of Vermont. **First author.**

Cuddihy, T. S. (2023). *Dissolved Oxygen Dynamics in Restored Wetlands in Vermont and the Implications for Phosphorus Cycling.* Undergraduate Honors Thesis, University of Vermont.

SKILLS

LANGUAGES & DATA

R · Python · geopandas · rasterio · GDAL · SciPy · SQL / SQLite

GEOSPATIAL & REMOTE SENSING

ArcGIS Pro · QGIS · Remote sensing · LiDAR (QTM) · Raster analysis · Spatial statistics

ENGINEERING & WORKFLOW

Git & GitHub · Data pipelines / ETL · Reproducible workflows

ANALYSIS

Statistical modeling · Compositional analysis · Data visualization

TOOLS

Adobe Suite · Microsoft Office · Google Workspace

EDUCATION

B.S. Environmental Science

Ecological Design · Honors College

University of Vermont · 2019–2023

GPA 3.63

Honors Thesis: dissolved-oxygen dynamics in the Lake Champlain Basin. View →